

TidyR_homework

Juelle St. Clair

2026-03-31

TidyR Homework

In this assignment, you will be asked to select a dataset of your choice, identify any ‘untidy’ characteristics, make it tidy, and do some exploratory analysis.

Question 1 (1 point)

Select a dataset from <https://github.com/rfordatascience/tidytuesday/tree/main/data> and import it into RStudio.

If your dataset references a paper, like the one we did in class, you can download it from its data repository. If it is only found on the tidy tuesday github, you may need to use one of these functions: `tidytuesdayR::tt_load` or `readr::read_csv`

Show me the first 10 rows of your data (1 point).

```
# Question 1
```

```
#load a dataset
```

```
edible_plants <- readr::read_csv('https://raw.githubusercontent.com/rfordatascience/tidytuesday/main/data/edible_plants.csv')
```

```
## Rows: 140 Columns: 20
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (17): taxonomic_name, common_name, cultivation, sunlight, water, nutrien...
```

```
## dbl (3): preferred_ph_lower, preferred_ph_upper, energy
```

```
##
```

```
## i Use 'spec()' to retrieve the full column specification for this data.
```

```
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
#output the first 10 rows of your data
```

```
?head
```

```
## starting httpd help server ... done
```

```
head(edible_plants, n = 10L)
```

```
## # A tibble: 10 x 20
##   taxonomic_name      common_name cultivation sunlight water preferred_ph_lower
##   <chr>              <chr>         <chr>      <chr>    <chr>          <dbl>
## 1 Pisum sativum      Pea           Legume     Full sun Very~          6
## 2 Phaseolus coccineus Beans (Run~ Legume     Full sun Medi~          6
## 3 Phaseolus vulgaris Beans (Fre~ Legume     Full sun Medi~          6
## 4 Vicia faba         Beans (Bro~ Legume     Full su~ Very~          6
## 5 Brassica oleracea   Cabbage (S~ Legume     Full sun High           6
## 6 Brassica oleracea ~ Broccoli Brassica     Full su~ Medi~          6
## 7 Brassica oleracea ~ Brussels S~ Brassica     Full sun Medi~          6
## 8 Brassica oleracea ~ Cauliflower Brassica     Full sun Medi~          5.5
## 9 Brassica oleracea ~ Kale         Brassica     Full sun Low            6
## 10 Brassica rapa (Rap~ Turnip      Brassica     Full su~ Medi~          5.5
## # i 14 more variables: preferred_ph_upper <dbl>, nutrients <chr>, soil <chr>,
## #   season <chr>, temperature_class <chr>, temperature_germination <chr>,
## #   temperature_growing <chr>, days_germination <chr>, days_harvest <chr>,
## #   nutritional_info <chr>, energy <dbl>, sensitivities <chr>,
## #   description <chr>, requirements <chr>
```

Question 2 (3 points)

Identify at least 3 non-tidy characteristics of your data or things you would like to change in this dataset (1 point ea):

- 1. change the column name 'water' to 'water requirements'
- 2. change the column name 'sunlight' to 'sunlight requirements'
- 3. change the column name 'nutrients' to 'nutrients requirements'
- 4. change the column name 'soil' to 'preferred soil type'
- 5. change the column name 'energy' to 'energy value'
- 6. change the column name 'season' to 'growing season'
- 7. change the column name 'requirements' to 'plant requirements'
- 8. rearrange column order

Show any code you used to identify these characteristics:

```
# Question 2
```

```
colnames(edible_plants)
```

```
## [1] "taxonomic_name"      "common_name"
## [3] "cultivation"         "sunlight"
## [5] "water"               "preferred_ph_lower"
## [7] "preferred_ph_upper"  "nutrients"
## [9] "soil"                "season"
## [11] "temperature_class"   "temperature_germination"
## [13] "temperature_growing" "days_germination"
## [15] "days_harvest"       "nutritional_info"
## [17] "energy"              "sensitivities"
## [19] "description"         "requirements"
```

Question 3 (5 points)

Fix each of your issues. Use comments to show where each of your issues listed above was fixed. Use pipes to connect fixing all of your issues together!

```
# Question 3

edible_plants_cleaned <- edible_plants %>%
  rename(
    `water requirements` = water,           # renamed all the columns and fixed problems 1-7
    `sunlight requirements` = sunlight,
    `nutrient requirements` = nutrients,
    `preferred soil type` = soil,
    `energy value` = energy,
    `growing season` = season,
    `plant requirements` = requirements
  ) %>%
  relocate (taxonomic_name,
            common_name,                 # rearranged a few columns in the data base
            cultivation,
            `water requirements`,
            `sunlight requirements`,
            temperature_germination,
            temperature_growing,
            temperature_class,
            sensitivities
  )
```

Question 4 (2 points)

List a question you would like to answer using this dataset:

Which cultivation has the most edible plants?

Question 5 (5 points)

Using stats, plots, or both, answer your question. Make sure you output any stats or plots you create in this code chunk (i.e., that it shows up in the knitted document).

NOTE: you do not have to use a formal stats test if you are not familiar! Feel free to just use summary stats and/or figures to answer your question.

```
# Question 5

cultivations <- edible_plants_cleaned %>%
  group_by(cultivation)%>%
  summarise(species = n())

cultivations
```

```
## # A tibble: 12 x 2
##   cultivation    species
##   <chr>         <int>
```

##	1	Allium	9
##	2	Brassica	21
##	3	Brassicas	1
##	4	Chenopodiaceae	3
##	5	Cucurbit	8
##	6	Lamiaceae	4
##	7	Legume	10
##	8	Miscellaneous	65
##	9	Salad	3
##	10	Solanaceae	6
##	11	Solanum	2
##	12	Umbelliferae	8

Provide the answer (in words) to your question below:

The cultivation Miscellaneous has the most edible plants species according to the dataset

Question 6 (4 points)

Find two functions we haven't introduced you to that are a part of tidyR, dplyr, or other related tidyverse packages. Provide the function name (including the package it's from) and what it does.

The function `separate()` from `tidyr` is used to split a single column into multiple columns based on a delimiter

The function `case_when()` from `dplyr` is used to create a new column (or modify one) using multiple conditional statements like `if` or `else`